Patterns

25 March 2026 20:13

Q.
```
  *
 ***
*****
 ***
  *
```

Q.
```
*****
 ***
  *
 ***
*****
```

Q. Convert binary number to decimal.

Q. Prime factorization.

multiblier

- 1 1 0 0 1 1
  5 4 3 2 1 0

$1\times2^0 + 1\times2^1 + 0\times2^2 + 0\times2^3 + 1\times2^4 + 1\times2^5$

$= 51$

int count = 0;
while (num > 0)

```
int Count=0;    int ans = 0;
while (num > 0){
   int multiblier = num%10;
   ans = ans + multiblier * Math.pow (2, count);
   count ++;
}
```

1 1 0 0 1 1
5 4 3 2 1 0

count = 0

$1\times2^0 + 1\times2^1 + 0\times2^2$

1 1 0 0 1

count = 1

count = 2

$(Base)^{Power}$

Math.Pow(Base, Power)

2 count

Count = 0

$2^0, 2^1, 2^2$

$\boxed{\text{Math.Pow( )}}$ → use

$\boxed{\text{Math.Pow( )}} \rightarrow$ use ↓

↳ Slow → algorithm ↑

↳ backend (log complex)

```
Scanner sc = new Scanner(System.in);
    System.out.print("Enter binary number: ");
    int binary = sc.nextInt();
    int decimal = 0, base = 1, rem;
    while (binary > 0) {
        rem = binary % 10;
        decimal = decimal + rem * base;
        base = base * 2;
        binary = binary / 10;
    }
    System.out.println("Decimal = " + decimal);
>
```

2^0

multiplier

ans

1 1 0 0 1 1
5 4 3 2 1 0

1 → 2 → 4 → 8

$= 1\times 2^0 + 1\times 2^1 + 0\times 2^2 + 0\times 2^3 + 1\times 2^4 + 1\times 2^5$

1 $\xrightarrow{\times 2}$ 2 $\xrightarrow{\times 2}$ 4 $\xrightarrow{\times 2}$ 8 $\xrightarrow{\times 2}$ 16 32

Math · Pow( );

$2^0 = 1$

↳ 1

base = 1

1×1 $\boxed{\text{ans} = 1}$

ans = ans + multiplier × base

base = base × 2;

$2^3 = 2\times 2\times 2$

$1\times 2\times 2$

$= 4$

Count = 0

$\boxed{2^0 = 1}$ →

Count = 1

↳ Pow (2,1)

$\boxed{2^1}$

Count: 2

Pow (2,2)

$\boxed{2^2}$

base = 1 $\xrightarrow{\times 2}$ 2 $\xrightarrow{\times 2}$ 4

$\boxed{2^1 \times 2 = 2^2}$

$\boxed{16 \rightarrow 1, 2, 4, 8, 16}$

$16 \rightarrow 2 \times 2 \times 2 \times 2 = 16$

$1 \rightarrow$ Not a prime

$\rightarrow$ Neither Composite

$28 \Rightarrow 2 \times 2 \times 7 = 28$

$16 \rightarrow$ $\boxed{2 \rightarrow 16}$

$16/6$ ✗ $16/5$ ✗ $16/4$ ✗ 2, 3 $16/3$ ✗

num = 16 → 1

$\frac{16}{2} = \frac{8}{2} \rightarrow \frac{8}{2} \rightarrow \frac{4}{2} \rightarrow \frac{2}{2} \rightarrow \frac{1}{2} \rightarrow$ ✗

$\rightarrow$ 2 2 2 2

28

$\rightarrow (2 \rightarrow 28)$ ✓

$\rightarrow \frac{28}{2} \rightarrow \frac{14}{2} \rightarrow \frac{7}{2}$ ✗

num = 7

$\frac{7}{3}$ ✗

2 2 7

$\frac{7}{4}$ ✗

$\frac{7}{5}$ ✗

$\boxed{\text{num = ip();}}$

$\frac{7}{6}$ ✗ $\frac{7}{7}$ 1 num = 1;

```
int n=num;
for(int i=2; i<=num; i++){
```

$\frac{16}{2} = \frac{8}{2}$

```
for(int i=2; i<=num; i++){
    while(n % i ==0){
        Sout(i);
        n=n/i;
    }
```

$\frac{16}{2} = \frac{8}{2}$

2 2 2

$\frac{8}{2} = \frac{4}{2} = \frac{2}{2} = 1$

i=2 +1 = 3

$\boxed{n=1}$ 1/3

i =4

254

~~264~~

127

2

$\sqrt{n}$

36 → 1, 36
2, 18
- -
(6, 6)

$\sqrt{254}$

= 15

15 : 9
↳ 15

$-\infty$ to ∞

-1 0 1 2 3

2 → 15

2 2 7

n=

n=28 $\sqrt{28}$ = 5

2 . 2 7

28

$\frac{28}{2}$

$\frac{14}{2}$ = 7

(2 → 5)

2, 3, 4, 5

2 · 2 7

n = 7

Already prime

[1 — $\sqrt{n}$]

7

n ·

(1, 36)

(2, 18)

(3, 12)

(4, 9)

~~(6, 6)~~

(9, 4)

36

1 — 6

$\frac{36}{4} = 9$

$\sqrt{n}$ >